

mdstats 0.20.194a0 - CUEQ-DEFAULT1 hotfix

Scope

This release fixes two workstation failures discovered immediately after enabling the phase-separated CuEq TRAIN2 default in 0.20.193a0.

Fix 1 - foundation contract reporting

Revision 60 correctly migrated the foundation configuration contract from one broad backend field to independent source_backend and training_backend fields. The doctor display path still indexed the retired backend key and therefore emitted foundation configuration is invalid: 'backend'. The display now reports both frozen phase choices directly. No scientific configuration changes are made by this fix.

Fix 2 - selected-head TRAIN2 CuEq roundoff policy

The generic ACCEL1 source/DATA6 FP32 parity policy remains $rtol=1e-5$, $atol=1e-6$. It is not relaxed.

The EXTRACT1-derived selected-head checkpoint is a different execution authority used only as the TRAIN2 starting model. The workstation evidence that originally motivated CUEQ-PHASE1 already measured approximately $F_{max}=1.669e-6$ and $D_{max}=1.192e-6$ with identical deterministic selection. Therefore revision 61 freezes a distinct TRAIN2 preflight parity policy at $rtol=1e-5$, $atol=2e-6$ for FP32; FP64 remains $rtol=1e-10$, $atol=1e-12$.

The newly reported workstation values ($E_{max}=3.576e-7$, $F_{max}=1.490e-6$, $S_{max}=1.564e-8$, $D_{max}=1.570e-6$, identical selection) are inside that previously motivated TRAIN2 roundoff envelope. This does not authorize CuEq source inference or DATA6 and does not affect the original six-head source-parity gate.

Fail-closed behavior

CuEq TRAIN2 remains fail-closed for unavailable runtime components, non-finite output, selection drift, or numerical discrepancies outside the dedicated TRAIN2 policy. No silent e3nn fallback is introduced.